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TISSUE EXTRACTION DEVICES AND METHODS

Abstract

A tissue cutting device has an outer sleeve with a distal window and an inner cutting sleeve which moves past the window to cut tissue. The inner cutting sleeve has a lumen which may have a larger proximal diameter than distal diameter. A perimeter of the window may comprise a dielectric material. A distal edge of the inner sleeve may be displaced inwardly.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 18/496,478, filed Oct. 27, 2023, which is a continuation of U.S. application Ser. No. 15/801,687, filed Nov. 2, 2017, now U.S. Pat. No. 11,832,874, which is a continuation of U.S. application Ser. No. 14/982,593, filed Dec. 29, 2015, now U.S. Pat. No. 9,827,037, which is a continuation of U.S. application Ser. No. 13/442,686, filed Apr. 9, 2012, now U.S. Pat. No. 9,254,142, which claims the benefit of U.S. Provisional Application No. 61/474,164, filed Apr. 11, 2011, and U.S. Provisional Application No. 61/501,438, filed Jun. 27, 2011. This application is also related to but does not claim priority from U.S. application Ser. No. 13/277,913, filed Oct. 20, 2011, and U.S. application Ser. No. 13/287,315, filed Nov. 2, 2011. The entire contents of each of these applications is incorporated herein in its entirety by reference.

TECHNICAL FIELD

[0002] The present invention relates systems and methods for the cutting and extraction of uterine fibroid tissue, polyps and other abnormal uterine tissue.

BACKGROUND

[0003] Uterine fibroids are non-cancerous tumors that develop in the wall of uterus. Such fibroids occur in a large percentage of the female population, with some studies indicating up to 40 percent of all women have fibroids. Uterine fibroids can grow over time to be several centimeters in diameter and symptoms can include menorrhagia, reproductive dysfunction, pelvic pressure and pain.

[0004] One current treatment of fibroids is hysteroscopic resection or myomectomy which involves transcervical access to the uterus with a hysteroscope together with insertion of a cutting instrument through a working channel in the hysteroscope. The cutting instrument may be a mechanical tissue cutter or an electrosurgical resection device such as a cutting loop. Mechanical cutting devices are disclosed in U.S. Pat. Nos. 7,226,459; 6,032,673 and 5,730,752 and U.S. Published Patent Appl. 2009/0270898. An electrosurgical cutting device is disclosed in U.S. Pat. No. 5,906,615.

[0005] In a myomectomy or hysteroscopic resection, the initial step of the procedure includes distention of the uterine cavity to create a working space for assisting viewing through the

hysteroscope. In a relaxed state, the uterine cavity collapses with the uterine walls in contact with one another. A fluid management system is used to distend the uterus to provide a working space wherein a fluid is administered through a passageway in the hysteroscope under sufficient pressure to expand or distend the uterine cavity. The fluids used to distend the uterus are typically liquid aqueous solutions such as a saline solution or a sugar-based aqueous solution.

[0006] In some RF electrosurgical resection procedures, the distending fluid is a non-conductive aqueous solution to limit RF current conduction.

[0007] One particular concern is the fact that fluid management systems typically administer the fluid under a pressure of up to 100 mm Hg or more which results in a significant risk that the distending fluid may be taken up by a cut blood vessel exposed in the uterine cavity. Such unwanted fluid uptake is known as intravasation, which can lead to serious complications and even death. For this reason, fluid management systems have been developed to monitor the patient's fluid uptake on a continuous basis during a procedure, typically using complicated systems that capture, collect and weigh distending fluids that flow through the uterine cavity.

[0008] While hysteroscopic resection can be effective in removing uterine fibroids, many commercially available instrument are too large in diameter and thus require anesthesia in an operating room environment. Conventional resectoscopes require cervical dilation to about 9 mm. What is needed is a system that can effectively cut and remove fibroid tissue through a small diameter hysteroscope.

BRIEF SUMMARY

[0009] The present invention comprises a tissue cutting device including an elongated structure where the elongated structure comprises an outer sleeve and an inner cutting sleeve. The elongated structure will typically be attached at a proximal end to a hub, handle, or other component to allow manipulation of the device. The elongated structure will be in the form of a shaft suitable for introduction to a body cavity, for example for transcervical introduction to a uterus. A cutting window will be formed through a wall at a distal end of the sleeve, and the inner sleeve will be configured to move between a proximal position and a distal position relative to the cutting window. In particular, the inner cutting sleeve will typically have a distal cutting edge, such as a sharpened edge, an electrosurgical edge, or the like, which allows the inner sleeve to be advanced past the cutting window in the outer sleeve in order to cut, sever, or otherwise remove tissue which intrudes inwardly through the window. The inner sleeve will have an interior lumen extending at least partly therethrough to allow tissue to be extracted after it has been severed.

[0010] In a first aspect of the present invention, the tissue-extracting lumen of the inner cutting sleeve has a distal luminal portion having a first diameter and a proximal luminal portion having a second diameter. The first distal diameter will usually be less than the second proximal diameter so that the cross-sectional area of the distal portion is less than that of the proximal portion of the lumen.

[0011] Usually, the distal lumen portion will have a length which is only a small fraction of the length of the total tissue-extracting lumen. The total length of the tissue-extracting lumen for a device configured for transcervical introduction into a uterus, including both the distal and proximal luminal portions, will typically be in the range from 450 mm to 550 mm, with the distal luminal portion with usually extending at least 1 mm from a distal end of the inner cutting sleeve, often extending at least 2 mm from the distal end of the inner cutting sleeve. The distal luminal portion will usually have a maximum length no greater than 15 mm, more usually being no greater than 10 mm. Thus, the distal lumen portion will usually have a length and range from 1 mm to 15 mm, often being in the range from 2 mm to 10 mm.

[0012] The inner cutting sleeve and the outer sleeve will typically both be cylindrical at at least their distal ends in order to permit relative rotation and optionally rotational oscillation. In other instances, however, the sleeves or some portion thereof could have non-circular cross-sectional areas where the diameter ranges set forth above will be equivalent to an average width of the lumen

in any point. The cross-sectional area of the distal luminal portion will typically be less than 95% of the cross-sectional area of the proximal luminal portion, more usually being less than 90% of the cross-sectional area of the proximal luminal portion. Usually, the cross-sectional area of the distal luminal portion will be less than 95% of the cross-sectional area of the proximal luminal portion, more usually being less than 90% of the cross-sectional area of the proximal luminal portion. The cross-sectional area of the distal luminal portion relative to the proximal luminal portion will typically be in the range from 80% to 95%, more usually in the range from 90% to 95%.

[0013] In another aspect of the present invention, a perimeter of the distal or cutting window in the outer sleeve will comprise a dielectric material, such as a ceramic material, a polymeric material, or the like. The dielectric material about the window perimeter functions to provide spacing between the opposing polarity electrode surfaces of the inner cutting sleeve and the outer sleeve.

[0014] In a further aspect of the present invention, the inner sleeve of the tissue cutting device has a distal tissue-contacting edge that is displaced radially inwardly from an outer diameter of said inner sleeve, where the displaced edge typically comprises an electrode. In such cases, a dielectric layer will usually be formed on or over an interior of the outer sleeve.

[0015] In a still further aspect of the present invention, a ratio of the diameter of at least a proximal portion of the tissue-extraction lumen to the outer diameter of the outer sleeve will be at least 0.65 to 1, usually being at least 0.70 to 1.

[0016] In yet another aspect of the present invention, the tissue cutting system will further comprise a hysteroscope, and a ratio of the diameter of the tissue-extraction lumen to an outer diameter of the hysteroscope will be at least 0.35 to 1, preferably being at least 0.4 to 1.

[0017] In still another aspect of the present invention, a tissue cutting probe comprises first and second concentric sleeves having a common axis, where the sleeves are configured for relative axial movement between a window-open position for receiving tissue therethrough and a range of window-closing positions in which the second sleeve is adapted to cut tissue which is received through the window in the first sleeve. The probe further comprises a motor to provide relative movement of the sleeves, including rotation, rotational oscillation, and/or axial translation) to effect cutting, and each of the sleeves comprises a tissue-contacting edge, where the contacting edge of one sleeves comprises an electrode and the contacting edge of the other sleeve comprises a dielectric. The dielectric maybe on at least one interfacing surface of the first and second sleeves, and often will be on the surfaces of both the first and second sleeves. Optionally, the first and second sleeves comprise opposing polarity electrodes which maybe coupled to a radiofrequency source.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0018] FIG. 1 is a plan view of an assembly including a hysteroscope and a tissue-cutting device corresponding to the invention that is inserted through the working channel of the hysteroscope.

[0019] FIG. 2 is a schematic perspective view of a fluid management system used for distending the uterus and for assisting in electrosurgical tissue cutting and extraction.

[0020] FIG. 3 is a cross-sectional view of the shaft of the hysteroscope of FIG. 1 showing various channels therein.

[0021] FIG. 4 is a schematic view of the working end of the electrosurgical tissue-cutting device of FIG. 1 showing an outer sleeve with a reciprocating inner cutting sleeve in a partially advanced position.

[0022] FIG. 5 is a schematic perspective view of the working end of the inner sleeve of FIG. 4 showing its electrode edge.

[0023] FIG. 6A is a schematic cut-away view of a portion of outer sleeve, inner RF cutting sleeve

and a tissue-receiving window of the outer sleeve.

[0024] FIG. 6B is a schematic view of a distal end portion another embodiment of inner RF cutting sleeve.

[0025] FIG. 7A is a cross sectional view of the inner RF cutting sleeve of FIG. 6B taken along line 7A-7A of FIG. 6B.

[0026] FIG. 7B is another cross sectional view of the inner RF cutting sleeve of FIG. 6B taken along line 7B-7B of FIG. 6B.

[0027] FIG. 8 is a schematic view of a distal end portion of another embodiment of inner RF cutting sleeve.

[0028] FIG. 9A is a cross sectional view of the RF cutting sleeve of FIG. 8 taken along line 9A-9A of FIG. 8.

[0029] FIG. 9B is a cross sectional view of the RF cutting sleeve of FIG. 8 taken along line 9B-9B of FIG. 8.

[0030] FIG. 10A is an enlarged sectional view of a working end with an RF cutting sleeve in a partially advanced position illustrating fluid volumes that comprise a fluidic pumping mechanism corresponding to the invention for displacement of captured tissue.

[0031] FIG. 10B is another enlarged sectional view similar to FIG. 10A with the RF cutting sleeve in a further advanced position showing how a captured fluid volumes applies fluidic or hydraulic pressure to captured tissue.

[0032] FIG. 11A is a longitudinal sectional view a working end illustrating an electrode edge cutting tissue.

[0033] FIG. 11B is a longitudinal sectional view similar to FIG. 11A illustrating the pumping function of the working end.

[0034] FIG. 12A is a cross-section of a sleeve assembly showing a fluid inflow lumen.

[0035] FIG. 12B is a cross-section of a sleeve assembly showing an alternative fluid inflow lumen.

[0036] FIG. 12C is a cross-section of a sleeve assembly showing an alternative fluid inflow lumen.

[0037] FIG. 13 is a sectional view of a variation of cutting sleeve with an inwardly displaced electrode edge.

[0038] FIG. 14 is a perspective view of another variation with an exterior sleeve with a distal dielectric body portion.

DETAILED DESCRIPTION

[0039] FIG. 1 illustrates an assembly that comprises an endoscope 50 used for hysteroscopy together with a tissue-extraction device 100 extending through a working channel 102 of the endoscope. The endoscope or hysteroscope 50 has a handle 104 coupled to an elongated shaft 105 having a diameter of 5 mm to 7 mm. The working channel 102 therein may be round, D-shaped or any other suitable shape. The endoscope shaft 105 is further configured with an optics channel 106 and one or more fluid inflow/outflow channels 108a, 108b (FIG. 3) that communicate with valve-connectors 110a, 110b configured for coupling to a fluid inflow source 120 thereto, or optionally a negative pressure source 125 (FIGS. 1-2). The fluid inflow source 120 is a component of a fluid management system 126 as is known in the art (FIG. 2) which comprises a fluid container 128 and pump mechanism 130 which pumps fluid through the hysteroscope 50 into the uterine cavity. As can be seen in FIG. 2, the fluid management system 126 further includes the negative pressure source 125 (which can comprise an operating room wall suction source) coupled to the tissue-cutting device 100. The handle 104 of the endoscope includes the angled extension portion 132 with optics to which a videoscopic camera 135 can be operatively coupled. A light source 136 also is coupled to light coupling 138 on the handle of the hysteroscope 50. The working channel 102 of the hysteroscope is configured for insertion and manipulation of the tissue-cutting and extracting device 100, for example to treat and remove fibroid tissue. In one embodiment, the hysteroscope shaft 105 has an axial length of 21 cm, and can comprise a 0° scope, or 15° to 30° scope.

[0040] Still referring to FIG. 1, the tissue-cutting device 100 has a highly elongated shaft assembly

140 configured to extend through the working channel **102** in the hysteroscope. A handle **142** of the tissue-cutting device **100** is adapted for manipulating the electrosurgical working end **145** of the device. In use, the handle **142** can be manipulated both rotationally and axially, for example, to orient the working end **145** to cut targeted fibroid tissue. The tissue-cutting device **100** has subsystems coupled to its handle **142** to enable electrosurgical cutting of targeted tissue. A radio frequency generator or RF source **150** and controller **155** are coupled to at least one RF electrode carried by the working end **145** as will be described in detail below. In one embodiment shown in FIG. 1, an electrical cable **156** and negative pressure source **125** are operatively coupled to a connector **158** in handle **142**. The electrical cable couples the RF source **150** to the electrosurgical working end **145**. The negative pressure source **125** communicates with a tissue-extraction channel **160** in the shaft assembly **140** of the tissue extraction device **100** (FIG. 4).

[0041] FIG. 1 further illustrates a seal housing **162** that carries a flexible seal **164** carried by the hysteroscope handle **104** for sealing the shaft **140** of the tissue-cutting device **100** in the working channel **102** to prevent distending fluid from escaping from a uterine cavity.

[0042] In one embodiment as shown in FIG. 1, the handle **142** of tissue-cutting device **100** includes a motor drive **165** for reciprocating or otherwise moving a cutting component of the electrosurgical working end **145** as will be described below. The handle **142** optionally includes one or more actuator buttons **166** for actuating the device. In another embodiment, a footswitch can be used to operate the device. In one embodiment, the system includes a switch or control mechanism to provide a plurality of reciprocation speeds, for example 1 Hz, 2 Hz, 3 Hz, 4 Hz and up to 8 Hz. Further, the system can include a mechanism for moving and locking the reciprocating cutting sleeve in a non-extended position and in an extended position. Further, the system can include a mechanism for actuating a single reciprocating stroke.

[0043] Referring to FIGS. 1 and 4, an electrosurgical tissue-cutting device has an elongate shaft assembly **140** extending about longitudinal axis **168** comprising an exterior or first outer sleeve **170** with passageway or lumen **172** therein that accommodates a second or inner sleeve **175** that can reciprocate (and optionally rotate or oscillate) in lumen **172** to cut tissue as is known in that art of such tubular cutters. In one embodiment, the tissue-receiving window **176** in the outer sleeve **170** has an axial length ranging between 10 mm and 30 mm and extends in a radial angle about outer sleeve **170** from about 45° to 210° relative to axis **168** of the sleeve. In one embodiment, the window extends in a radial angle of to 180°—that is, the window cut is one-half of the tube. The outer and inner sleeves **170** and **175** can comprise a thin-wall stainless steel material and function as opposing polarity electrodes as will be described in detail below. FIGS. 6A-8 illustrate insulative or dielectric layers carried by the outer and inner sleeves **170** and **175** to limit, control and/or prevent unwanted electrical current flows between certain portions of the sleeve. In one embodiment, a stainless steel outer sleeve **170** has an O.D. of 0.143" with an I.D. of 0.133" and with an inner insulative layer (described below) the sleeve has a nominal I.D. of 0.125". In this embodiment, the stainless steel inner sleeve **175** has an O.D. of 0.112" with an I.D. of 0.106". The inner sleeve **175** with an outer insulative layer **202** has a nominal O.D. of about 0.120 to reciprocate in lumen **172**. In other embodiments, outer and or inner sleeves can be fabricated of metal, plastic, ceramic or a combination thereof. The cross-section of the sleeves can be round, oval or any other suitable shape.

[0044] In one embodiment, the outer sleeve is 17-7 PH stainless steel has an O.D. of 0.140 to 0.143" with a wall thickness of 0.005" to 0.007" and the inner sleeve also is 17-7. It can be understood that having the largest possible diameter extraction lumen **160** (FIG. 5) is advantageous, but limited by the O.D. of the shaft assembly which in turn is limited by the desired cross section of the hysteroscope **50**. To minimize dilation of the patient's cervix, the maximum scope diameter should be about 6.5 mm (0.256") which generally may allow for a maximum working channel of about 0.150". In one aspect of the invention, the thin wall tubing and insulation layers have been developed to provide an optimized tissue extraction lumen diameter (given the

above scope dimensions and limitations above) that is greater than 0.090" or greater than 0.100"—all accommodated in hysteroscope having a O.D. of 6.5 mm.

[0045] Thus, in general, a tissue cutting device corresponding to the invention comprises an elongated assembly comprising concentric outer and inner sleeves extending along an axis, a tissue-receiving window in the outer sleeve and a reciprocating inner sleeve having an extraction lumen **160** therein (FIG. 5), and wherein the ratio of the diameter of the extraction lumen **160** to the outer diameter of the outer sleeve is at least 0.65:1 or at least 0.70:1.

[0046] In another aspect, the tissue cutting device and hysteroscope together corresponding to the invention comprises an assembly or combination wherein the ratio of the diameter of the extraction lumen **160** to the outer diameter of the hysteroscope to at least 0.35:1 or at least 0.40:1.

[0047] As can be seen in FIG. 4, the distal end **177** of inner sleeve **175** comprises a first polarity electrode with distal cutting electrode edge **180** about which plasma can be generated. The electrode edge **180** also can be described as an active electrode during tissue cutting since the electrode edge **180** then has a substantially smaller surface area than the opposing polarity or return electrode. In one embodiment in FIG. 4, the exposed surfaces of outer sleeve **170** comprises the second polarity electrode **185**, which thus can be described as the return electrode since during use such an electrode surface has a substantially larger surface area compared to the functionally exposed surface area of the active electrode edge **180**.

[0048] In one aspect of the invention, the inner sleeve or cutting sleeve **175** has an interior tissue extraction lumen **160** with first and second interior diameters that are adapted to electrosurgically cut tissue volumes rapidly—and thereafter consistently extract the cut tissue strips through the highly elongated lumen **160** without clogging. Now referring to FIGS. 5 and 6A, it can be seen that the inner sleeve **175** has a first diameter portion **190A** that extends from the handle **142** (FIG. 1) to a distal region **192** of the sleeve **175** wherein the tissue extraction lumen transitions to a smaller second diameter lumen **190B** with a reduced diameter indicated at B which is defined by the electrode sleeve element **195** that provides cutting electrode edge **180**. The axial length C of the reduced cross-section lumen **190B** can range from about 1 mm to 15 mm. In one embodiment, the first diameter A is 0.106" and the second reduced diameter B is 0.095" and has an axial length of 2 mm. In one embodiment, the cross-sectional area of the distal lumen portion is less than 95% of cross-sectional area of the proximal lumen portion, or less than 90% of the cross-sectional area of the proximal lumen portion. As shown in FIG. 5, the inner sleeve **175** can be an electrically conductive stainless steel and the reduced diameter electrode portion also can comprise a stainless steel electrode sleeve element **195** that is welded in place by weld **196** (FIG. 6A). In another alternative embodiment, the electrode and reduced diameter electrode sleeve element **195** comprises a tungsten tube that can be press fit into the distal end **198** of inner sleeve **175**. FIGS. 5 and 6A further illustrates the interfacing insulation layers **202** and **204** carried by the first and second sleeves **170**, **175**, respectively. In FIG. 6A, the outer sleeve **170** is lined with a thin-wall insulative material **200**, such as PFA, or another material described below. Similarly, the inner sleeve **175** has an exterior insulative layer **202**. These coating materials can be lubricious as well as electrically insulative to reduce friction during reciprocation of the inner sleeve **175**.

[0049] The insulative layers **200** and **202** described above can comprise a lubricious, hydrophobic or hydrophilic polymeric material. For example, the material can comprise a bio-compatible material such as PFA, TEFLON®, polytetrafluoroethylene (PTFE), FEP (Fluorinated ethylenepropylene), polyethylene, polyamide, ECTFE (Ethylenechlorotrifluoro-ethylene), ETFE, PVDF, polyvinyl chloride or silicone.

[0050] Now turning to FIG. 6B, another variation of inner sleeve **175** is illustrated in a schematic view together with a tissue volume being resected with the plasma electrode edge **180**. In this embodiment, as in other embodiments in this disclosure, the RF source operates at selected operational parameters to create a plasma around the electrode edge **180** of electrode sleeve **195** as is known in the art. Thus, the plasma generated at electrode edge **180** can cut and ablate a path P in

the tissue **220**, and is suited for cutting fibroid tissue and other abnormal uterine tissue. In FIG. **6B**, the distal portion of the cutting sleeve **175** includes a ceramic collar **222** which is adjacent the distal edge **180** of the electrode sleeve **195**. The ceramic **222** collar functions to confine plasma formation about the distal electrode edge **180** and functions further to prevent plasma from contacting and damaging the polymer insulative layer **202** on the cutting sleeve **175** during operation. In one aspect of the invention, the path P cut in the tissue **220** with the plasma at electrode edge **180** provides a path P having an ablated width indicated at W, wherein such path width W is substantially wide due to tissue vaporization. This removal and vaporization of tissue in path P is substantially different than the effect of cutting similar tissue with a sharp blade edge, as in various prior art devices. A sharp blade edge can divide tissue (without cauterization) but applies mechanical force to the tissue and may prevent a large cross section slug of tissue from being cut. In contrast, the plasma at the electrode edge **180** can vaporize a path P in tissue without applying any substantial force on the tissue to thus cut larger cross-sections of strips of tissue. Further, the plasma cutting effect reduces the cross section of tissue strip **225** received in the tissue-extraction lumen **190B**. FIG. **6B** depicts a tissue strip to **225** entering lumen **190B** which has such a smaller cross-section than the lumen due to the vaporization of tissue. Further, the cross section of tissue **225** as it enters the larger cross-section lumen **190A** results in even greater free space **196** around the tissue strip **225**. Thus, the resection of tissue with the plasma electrode edge **180**, together with the lumen transition from the smaller cross-section (**190B**) to the larger cross-section (**190A**) of the tissue-extraction lumen **160** can significantly reduce or eliminate the potential for successive resected tissue strips **225** to clog the lumen. Prior art resection devices with such small diameter tissue-extraction lumen typically have problems with tissue clogging.

[0051] In another aspect of the invention, the negative pressure source **225** coupled to the proximal end of tissue-extraction lumen **160** (see FIGS. **1** and **4A**) also assists in aspirating and moving tissue strips **225** in the proximal direction to a collection reservoir (not shown) outside the handle **142** of the device.

[0052] FIGS. **7A-7B** illustrate the change in lumen diameter of cutting sleeve **175** of FIG. **6B**. FIG. **8** illustrates the distal end of a variation of cutting sleeve **175'** which is configured with an electrode cutting element **195'** that is partially tubular in contrast to the previously described tubular electrode element **195** (FIGS. **5** and **6A**). FIGS. **9A-9B** again illustrate the change in cross-section of the tissue-extraction lumen between reduced cross-section region **190B'** and the increased cross-section region **190A'** of the cutting sleeve **175'** of FIG. **8**. Thus, the functionality remains the same whether the cutting electrode element **195'** is tubular or partly tubular. In FIG. **8A**, the ceramic collar **222'** is shown, in one variation, as extending only partially around sleeve **175** to cooperate with the radial angle of cutting electrode element **195'**. Further, the variation of FIG. **8** illustrates that the ceramic collar **222'** has a larger outside diameter than insulative layer **202**. Thus, friction may be reduced since the short axial length of the ceramic collar **222'** interfaces and slides against the interfacing insulative layer **200** about the inner surface of lumen **172** of outer sleeve **170**.

[0053] In general, one aspect of the invention comprises a tissue cutting and extracting device (FIGS. **4A-4B**) that includes first and second concentric sleeves having an axis and wherein the second (inner) sleeve **175** has an axially-extending tissue-extraction lumen therein, and wherein the second sleeve **175** is moveable between axially non-extended and extended positions relative to a tissue-receiving window **176** in first sleeve **170** to resect tissue, and wherein the tissue extraction lumen **160** has first and second cross-sections. The second sleeve **175** has a distal end configured as a plasma electrode edge **180** to resect tissue disposed in tissue-receiving window **176** of the first sleeve **170**. Further, the distal end of the second sleeve, and more particularly, the electrode edge **180** is configured for plasma ablation of a substantially wide path in the tissue. In general, the tissue-extraction device is configured with a tissue extraction lumen **160** having a distal end portion with a reduced cross-section that is smaller than a cross-section of medial and proximal portions of

the lumen **160**.

[0054] In one aspect of the invention, referring to FIGS. **7A-7B** and **9A-9B**, the tissue-extraction lumen **160** has a reduced cross-sectional area in lumen region **190A** proximate the plasma cutting tip or electrode edge **180** wherein said reduced cross section is less than 95%, 90%, 85% or 80% than the cross sectional area of medial and proximal portions **190B** of the tissue-extraction lumen, and wherein the axial length of the tissue-extraction lumen is from 450 mm to 550 mm for access to a uterine cavity. In one embodiment of tissue-cutting device **100** for hysteroscopic fibroid cutting and extraction (FIG. **1**), the shaft assembly **140** of the tissue-cutting device is 35 cm in length. Devices used for other procedures can have tissue-extraction lumen that are at least 10 cm, 20 cm, 30 cm or 40 cm in length.

[0055] Now referring to FIG. **4**, FIGS. **10A-10B** and FIGS. **11A-11B**, one aspect of the invention comprises a “tissue displacement” or pump means that is configured to displace and move tissue strips **225** (see FIGS. **11A-11B**) in the proximal direction in lumen **160** of cutting sleeve **175** to thus ensure that tissue does not clog the lumen of the inner sleeve **175**. As can be seen in FIGS. **4** and **10A-10B**, the pump means as a tissue displacement mechanism comprises a volume of captured fluid indicated at **400** (phantom outline in FIG. **4**) that is captured in a terminal chamber **405** of the assembly that is defined by the lumen **172** of outer sleeve and the distal tip or body **232** that is fixedly attached to outer sleeve **170**.

[0056] As shown in FIG. **10A**, the proximal end **406** of the terminal chamber **405** is defined by a plane transverse to axis **168** at the distal edge of the window **176**. In one embodiment, the axial length L of the terminal chamber **405** is at least 3 mm, 4 mm or 5 mm. In general, a tissue cutting device comprises an elongated assembly comprising concentric outer and inner sleeves extending along an axis, a tissue-receiving window in the outer sleeve open to an interior lumen that extends to a terminal chamber that is distal to the window, wherein the terminal chamber defines a fluid volume of at least 0.01 mL, at least 0.02 mL or at least 0.04 mL. In one embodiment, the terminal chamber is cylindrical and has a length to diameter ratio of at least 1:1, or at least 1.5:1.

[0057] In one embodiment depicted in FIGS. **4** and **10A**, the fluid in chamber **405** functions as a fluid piston to pump fluid or tissue in cylindrical chamber **410** defined by the electrode sleeve **195** and the adjoining extraction lumen **160**. Of particular interest, the captured fluid **400** can function as a pump and can push a captured tissue strip **225** in the proximal direction from the small cross-section lumen **190B** in electrode sleeve **155** as the cutting or inner sleeve **175** moves to its fully advanced or extended position (see. FIG. **10B**). For this reason, the above described length L of the terminal chamber **405** is at least as great as the axial length E of the small cross-section lumen **190B** in the cutting sleeve **175**. Further, as depicted in FIGS. **10A-10B** and **11A-11B**, the pumping stroke Y of the cutting sleeve **175** extends at least about 3 mm, 4 mm or 5 mm distally beyond the distal edge of the window **290**. In another aspect, the stroke Y of the cutting sleeve **175** is at least 5% or 10% of the total stroke of the cutting sleeve (stroke X+stroke Y in FIG. **11A**).

[0058] In general, a method of cutting tissue corresponding to the invention comprises cutting tissue with a reciprocating cutting sleeve having an extending stroke and a retracting stroke within an outer sleeve, wherein the extending stroke cuts and captures tissue received by a tissue-receiving window in the outer sleeve; and moving the captured tissue proximally in the cutting sleeve with fluidic pressure. The fluidic pressure is provided by the captured fluid volume **400** in chamber **405** in reaction to the relative motion of the fluid volume **400** and the cutting sleeve **175** within which a tissue strip **225** is disposed. As can be understood from FIGS. **10B** and **11B**, the fluidic pressure is substantially applied when the extending stroke of the sleeve **175** is distal of the window. In one method, the fluidic pressure is applied by a volume of distending fluid from the fluid-immersed working space that is captured in the closed-end terminal chamber **405** of outer sleeve **170**. The method includes cutting tissue with an RF plasma at a distal electrode edge **180** of the electrode sleeve **195**. In another method, the fluid could be entirely or a partially supplied by a pressurized fluid inflow from a remote source through a flow passageway **415** in outer sleeve **170** as depicted

in FIG. 12A, a flow passageway **416** in inner sleeve **175** as depicted in FIG. 12B, or a flow passageway **420** formed intermediate the walls of the outer and inner sleeves **170**, **175** as depicted in FIG. 12C.

[0059] In one embodiment depicted in FIGS. **4** and **10A**, the fluid in chamber **405** functions as a fluid piston to pump fluid or tissue in cylindrical chamber **410** defined by the electrode sleeve **195** and the adjoining extraction lumen **160**. Of particular interest, the captured fluid **400** can function as a pump and can push a captured tissue strip **225** in the proximal direction from the small cross-section lumen **190B** in electrode sleeve **155** as the cutting or inner sleeve **175** moves to its fully advanced or extended position (see. FIG. **10B**). For this reason, the above described length L of the terminal chamber **405** is at least as great as the axial length E of the small cross-section lumen **190B** in the cutting sleeve **170**. Further, as depicted in FIGS. **10A-10B** and **11A-11B**, the stroke Y of the cutting sleeve **175** extends at least about 3 mm, 4 mm or 5 mm distally beyond the distal edge of the window **290**. In another aspect, the stroke Y of the cutting sleeve **175** is at least 5% or 10% of the total stroke of the cutting sleeve (stroke X+stroke Y in FIG. **11A**).

[0060] In general, a method of cutting tissue corresponding to the invention comprising cutting tissue with a reciprocating cutting sleeve having an extending stroke and a retracting stroke within an outer sleeve, wherein the extending stroke cuts and captures tissue received by a tissue-receiving window in the outer sleeve, and pushing the captured tissue in the proximal direction in the cutting sleeve with fluidic pressure provided by the captured fluid volume **400** in chamber **405**. Further, the fluidic pressure and pump or displacement mechanism is configured to push the captured tissue at least in part from a first smaller cross-section lumen **190B** to a second larger cross-section lumen **190A** in the cutting sleeve **175**. Thereafter, the negative pressure source can more effectively extract and aspirate the tissue from the lumen.

[0061] In another aspect and method of the invention, tissue is cut and extracted by (i) interfacing a probe working end with targeted tissue wherein the working end comprises an elongated outer sleeve with a window exposed to a reciprocating inner sleeve, (ii) extending the inner sleeve in a first cutting stroke distally across the window thereby cutting tissue disposed within the window and (iii) extending the inner sleeve in a second pumping stroke distally beyond the window thereby causing fluidic pressure to pump the tissue proximally in a lumen of the inner sleeve. The sequence can be repeated with the first cutting stroke, the second pumping stroke and the retracting stroke having a rate of at least 1 Hz, 2 Hz or 3 Hz. The tissue can be cut with an electrode edge or a blade edge.

[0062] In another variation, the terminal chamber **405** is configured to capture a fluid volume sufficient to fill said inner sleeve lumen **160** over a length of at least 3 mm, 4 mm or 5 mm.

[0063] In another variation, the cutting step can include applying RF current to generate plasma at an electrode edge **180** on inner sleeve **175** and further comprising the step of terminating RF current at the end of the first cutting stroke. Alternatively, the system and controller **155** can terminate RF current during the second cutting stroke. Alternatively, the controller **155** can terminate RF current during the retracting stroke.

[0064] In another variation, the controller can apply RF current to the electrodes during at least a portion of the retracting stroke to thereby cauterize adjacent tissue. The cautery effect can be provided during the retracting stroke at the same operational parameters as used during the first cutting stroke, or at different operational RF parameters than used during the first cutting stroke.

[0065] FIGS. **11A-11B** illustrate in more detail the functional pump aspects of the invention. In FIG. **11A**, the reciprocating cutting sleeve **175** is shown in a medial position advancing distally wherein plasma at the cutting electrode edge **180** is cutting a tissue strip **225** that is disposed within lumen **160** of the cutting sleeve **175**. The distending fluid (saline) **244** from the working space migrates through window **176** into terminal chamber **405** to provide the fluid volume that will be captured upon advancement of the cutting sleeve saline.

[0066] In FIG. **11A**, it can be seen that the system operates in first electrosurgical mode

corresponding to the reciprocation and axial range of motion of cutting sleeve **175** relative to the tissue-receiving window **176**. The electrical fields EF of the first mode are indicated in FIG. **11A**. As described above, the first RF mode can be used over an axial length of travel of inner cutting sleeve **175** as it crosses the tissue-receiving window **176**, or the over both the cutting and pumping strokes, and also optionally the retraction stroke.

[0067] FIG. **11B** illustrates the fluid volume **400** in chamber **405** functioning as a fluid piston and pumping fluid and thus hydraulic or fluidic pressure against tissue strip **225** and moving the tissue proximally relative to the chamber defined by electrode sleeve **195** and the adjoining extraction lumen **160** (cf. FIG. **10B**).

[0068] In another aspect of the invention, referring back to FIG. **4**, the working end comprises first and second converging tissue-contacting edges, **180** and **440**, for cutting tissue engaged by or proximate to such edges. In FIG. **4**, it can be seen that a first edge comprises the electrode edge **180** described previously. The second first edge **440** comprises an edge of the window **176** that comprises a dielectric or insulative material such as a ceramic or polymeric material. In one embodiment, the tissue cutting probe comprises first and second concentric sleeves having an axis and configured for relative axial movement between a window-open position for receiving tissue therein and a range of window-closing positions in which the second sleeve cuts tissue in a window in the first sleeve, a motor for providing relative movement of the sleeves, wherein each of said sleeves comprising a tissue-contacting edge for contacting tissue, and wherein the contacting edge of one sleeve comprising an electrode and the contacting edge of the other sleeve comprising a dielectric.

[0069] In one variation, the tissue cutting probe includes an electrically insulative layer disposed on at least one interfacing surface of the first and second sleeves. In another variation, the tissue cutting probe has an electrically insulative layer disposed on the interfacing surfaces of both the first and second sleeves.

[0070] In another variation, the interface between the first and second sleeves provides hydraulic resistance to substantially prevent liquid flow therethrough. To provide such hydraulic resistance the electrically insulative layer of one or both sleeves at the interface comprises a hydrophilic surface. In another variation, an electrically insulative layer comprises a hydrophobic surface or an ultrahydrophobic surface.

[0071] Now referring to FIG. **13**, in another feature of the invention, it has been found that controlled spacing between the plasma-forming electrode **180** and the return electrode **185** (FIG. **4**) is desirable. For this reason, the electrode-sleeve has a displaced tissue-contacting edge **445** that is displaced dimension DD radially inward from an outer diameter of said second or cutting sleeve **175**. In general, the tissue cutting probe comprises first and second concentric sleeves having an axis and configured for relative axial movement between a window-open position for receiving tissue therein and a range of window-closing positions in which the second sleeve cuts tissue in a tissue-receiving window in the first sleeve, a mechanism for providing relative movement of the sleeves, and wherein the second sleeve has a displaced tissue-contacting edge that is displaced selected dimension radially inward from an outer diameter of said second sleeve. In one variation, the edge **444** is displaced a dimension DD radially inward of at least 0.003". The inwardly spaced edge further can be configured with a ceramic collar fixed about the exterior of the sleeve proximate to the displaced edge (FIG. **13**).

[0072] In another aspect of the invention, referring to FIG. **14**, the working end **145'** can comprise an outer metal sleeve **170** that is coupled to a dielectric body **450** of a ceramic or polymeric material that comprises a thin-wall structure bonded to sleeve **170** and extends substantially around the perimeter of window **176** and provides the interior terminal fluid chamber described above. In another aspect of the invention, the dielectric material provides further spacing between the first polarity electrode surface of the inner cutting sleeve (see. FIG. **4**) and the second polarity electrode of the outer sleeve **170** (see FIGS. **4** and **14**). A lower portion **455** of the metal sleeve can remain in

place to add strength to the structure. In one variation, at least one aperture **460** is provided close to an axially-transverse plane at the distal edge of window **176**. Such an aperture can provide for rapid flow of distention fluid into the terminal chamber. Such at least one aperture will be blocked by sleeve **175** during the pumping stroke (see FIG. **11B**).

[0073] Although particular embodiments of the present invention have been described above in detail, it will be understood that this description is merely for purposes of illustration and the above description of the invention is not exhaustive. Specific features of the invention are shown in some drawings and not in others, and this is for convenience only and any feature may be combined with another in accordance with the invention. A number of variations and alternatives will be apparent to one having ordinary skills in the art. Such alternatives and variations are intended to be included within the scope of the claims. Particular features that are presented in dependent claims can be combined and fall within the scope of the invention. The invention also encompasses embodiments as if dependent claims were alternatively written in a multiple dependent claim format with reference to other independent claims.

Claims

1. A tissue resecting device comprising: an outer sleeve including a side window opening into a lumen of the outer sleeve, the outer sleeve defining a first electrode having a first polarity; an inner sleeve positioned in the lumen of the outer sleeve, the inner sleeve comprising: a tubular shaft having a tissue extraction lumen extending therethrough, and a second electrode secured to and extending distally from a distal end of the tubular shaft, the second electrode having a second polarity opposite the first electrode; a ceramic collar secured to an outer surface of the second electrode distal of the distal end of the inner sleeve; wherein the inner sleeve is configured to longitudinally reciprocate relative to the outer sleeve such that the second electrode is movable across the side window in a distal direction during a forward stroke and in a proximal direction during a retracting stroke; and a controller configured to apply RF current between the first electrode and the second electrode during at least a portion of the forward stroke.
2. The tissue resecting device of claim 1, wherein the outer surface of the second electrode is a convex outer surface.
3. The tissue resecting device of claim 1, wherein a proximal end of the second electrode is positioned proximal of the distal end of the tubular shaft.
4. The tissue resecting device of claim 2, wherein the proximal end of the second electrode extends into the tissue extraction lumen of the tubular shaft.
5. The tissue resecting device of claim 1, wherein the second electrode comprises a tissue-contacting edge configured to generate plasma when RF current is applied.
6. The tissue resecting device of claim 5, wherein the tissue-contacting edge of the second electrode is displaced radially inward from an outer diameter of the tubular shaft.
7. The tissue resecting device of claim 1, wherein the second electrode is welded to the distal end of the tubular shaft.
8. The tissue resecting device of claim 1, wherein the outer sleeve includes a dielectric material extending around a perimeter of the side window.
9. The tissue resecting device of claim 1, further comprising a dielectric layer on or over an interior surface of the outer sleeve.
10. The tissue resecting device of claim 1, wherein an inner diameter of the second electrode is less than a diameter of the tissue extraction lumen of the tubular shaft.
11. A tissue resecting device comprising: an outer sleeve including a side window opening into a lumen of the outer sleeve, the outer sleeve defining a first electrode having a first polarity; an inner sleeve positioned in the lumen of the outer sleeve, the inner sleeve comprising: a tubular shaft having a tissue extraction lumen extending therethrough, a second electrode secured to and

extending distally from a distal end of the tubular shaft, the second electrode having a second polarity opposite the first electrode; and a ceramic collar surrounding the second electrode distal of the distal end of the tubular shaft; wherein the inner sleeve is configured to reciprocate relative to the outer sleeve such that the second electrode is movable across the side window in a distal direction during a forward stroke and in a proximal direction during a retracting stroke; wherein the second electrode includes a tissue-contacting edge configured to generate plasma when RF current is applied.

12. The tissue resecting device of claim 11, wherein the tissue-contacting edge of the second electrode is displaced radially inward from an outer diameter of the tubular shaft.

13. The tissue resecting device of claim 11, wherein the second electrode is welded to the distal end of the tubular shaft.

14. The tissue resecting device of claim 11, wherein a proximal portion of the second electrode extends into the tissue extraction lumen of the tubular shaft.

15. The tissue resecting device of claim 11, wherein the outer sleeve includes a dielectric material extending around a perimeter of the side window.

16. A tissue resecting device comprising: an outer sleeve including a side window opening into a lumen of the outer sleeve, the outer sleeve defining a first electrode having a first polarity; an inner sleeve positioned in the lumen of the outer sleeve, the inner sleeve comprising: a tubular shaft having a tissue extraction lumen extending therethrough, a second electrode secured to and extending distally from a distal end of the tubular shaft, the second electrode having a second polarity opposite the first electrode, and a ceramic collar surrounding the second electrode distal of the distal end of the tubular shaft, wherein the ceramic collar is configured to confine plasma formation; wherein the tissue extraction lumen comprises: a distal portion defined by the second electrode having a first diameter, and a proximal portion defined by the tubular shaft having a second diameter greater than the first diameter, wherein a cross-sectional area of the distal portion is between 80% and 95% of a cross-sectional area of the proximal portion; wherein the inner sleeve is configured to reciprocate relative to the outer sleeve such that the second electrode is movable across the side window in a distal direction during a forward stroke and in a proximal direction during a retracting stroke.

17. The tissue resecting device of claim 16, wherein the second electrode includes a tissue-contacting edge configured to generate plasma when RF current is applied between the first and second electrodes.

18. The tissue resecting device of claim 16, wherein the second electrode is welded to the distal end of the tubular shaft.

19. The tissue resecting device of claim 16, wherein a proximal portion of the second electrode extends into the tissue extraction lumen of the tubular shaft.

20. The tissue resecting device of claim 16, wherein the outer sleeve includes a dielectric material extending around a perimeter of the side window.
